RUI ZHOU

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EDUCATION

- **The Hong Kong University of Science and Technology** *Sep. 2026 – Present*
  *Ph.D. in Artificial Intelligence, advised by **Prof. Sirui Han** and **Prof. Yike Guo*** Hong Kong, SAR
- **Wuhan University** *Sep. 2022 – Jun. 2026*
  B.Eng. in Electronic Information Engineering Wuhan, China
  - **GPA: 3.95**/4.00; **Rank: 3**/203 **(Top 1.5%)**

RESEARCH INTERESTS

My research focuses on **embodied intelligence** and **robot learning,** particularly leveraging **human demonstrations** for scalable and generalizable **dexterous and mobile manipulation**.

RESEARCH PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PREPRINTS

[C.1] **Rui Zhou***, Yibo Yuan*, Junkai Zhao*, Fangyuan Zhao, Xiaoguang Zhao, Shanghang Zhang†, Sirui Han† (2026). **DexRoam: Enabling Mobile Dexterous Manipulation from Egocentric Human Demonstrations**. *Conference on Robot Learning (CoRL) 2026*.

[C.2] **Rui Zhou**, Jingbin Liu†, Junbin Xie, Jianyu Zhang, Yingze Hu, Jiele Zhao (2026). **STAG-VIO: Stabilized Prompt-to-Geometry Interface for Robust Dynamic Visual–Inertial Odometry**. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2026*.

[C.3] Chengbo Yuan, **Rui Zhou***, Mengzhen Liu*, Yingdong Hu, Shengjie Wang, Li Yi, Chuan Wen, Shanghang Zhang, Yang Gao† (2025). **MotionTrans: Human VR Data Enable Motion-Level Learning for Robotic Manipulation Policies**. *IEEE International Conference on Robotics and Automation (ICRA) 2026*.

[C.4] Qinwen Xu*, Jiaming Liu*, **Rui Zhou***, Shaojun Shi*, Nuowei Han*, Zhuoyang Liu, Chenyang Gu, Shuo Gu, Yang Yue, Gao Huang, Wenzhao Zheng, Sirui Han, Peng Jia, Shanghang Zhang† (2026). **TwinRL: Digital Twin-Driven Reinforcement Learning for Real-World Robotic Manipulation**. *ACM Multimedia (MM) 2026*.

[P.1] Yueru Jia*, Jiaming Liu*, Shengbang Liu*, **Rui Zhou**, Wanhe Yu, Yuyang Yan, Xiaowei Chi, Yandong Guo, Boxin Shi, Shanghang Zhang† (2025). **Video2Act: A Dual-System Video Diffusion Policy with Robotic Spatio-Motional Modeling**. *arXiv:2512.03044*.

EXPERIENCE

- **Beijing Academy of Artificial Intelligence[]** *Sep. 2025 - Sep. 2026*
  *Research Intern, Supervised by **Prof. Shanghang Zhang*** Beijing, China
  - Developed **DexRoam**, a complete system for learning mobile bimanual dexterous manipulation from egocentric whole-body human demonstrations, integrating portable capture, explicit whole-body alignment, and real-robot policy learning under a single *continuous, coupled* action interface
  - Built a **tracker-free** egocentric whole-body capture system recording full-body motion, hand articulation, and stereo video with only a consumer VR headset and a head-mounted camera, with in-headset streaming that keeps the demonstrator's view consistent with the policy observation
  - Designed three explicit alignment stages (**embodiment, action-semantic, temporal**) mapping human motion into a unified action space while preserving continuity, making human and robot data interchangeable for VLA policies
  - Validated on 5 real-world tasks and 2 VLA backbones: raised average success from **29%**→**56%** on GR00T N1.7 and **32%**→**57%** on $\pi_{0.5}$, matching robot-only training with **50% fewer** robot demonstrations
- **EVAR Lab, IIIS, Tsinghua University []** *Dec. 2024 - Aug. 2025*
  *Research Intern, Supervised by **Prof. Yang Gao*** Beijing, China
  - Systematically explored whether human data can enable robot policies to directly learn new motions for task completion, through multi-task human–robot cotraining
  - We propose **MotionTrans**, a framework that enables motion-level learning from VR-collected human data, allowing end-to-end robot policies to directly acquire new motions for task completion.
  - Built the Quest 3 teleoperation system for both the real robot (Franka arm + Inspire hand) and bimanual Isaac Gym simulation, open-sourced as **DexTeleGym[]**, featuring low-latency WebRTC streaming, real-time IK, dexterous hand retargeting, and gesture-controlled episode recording
  - By cotraining 30 human–robot tasks simultaneously, directly transferred motions of **13** tasks from human data to deployable end-to-end robot policies, with **9** tasks achieving non-trivial success rates in a **zero-shot** manner, and enhanced pretraining–finetuning performance by **+40%** success rate

- **HMI Lab, School of Computer Science, Peking University [🌐]** *Aug. 2025 - Feb. 2026*
  *Research Intern, Supervised by **Prof. Shanghang Zhang*** Beijing, China
  - We propose **TwinRL**, a digital twin–real-world collaborative post-training framework that expands and guides RL exploration for VLA models through SFT warm-up, twin RL warm-up, and real-world RL
  - Conducted real-world human-in-the-loop online RL experiments on a 7-DoF Franka FR3; across four tasks, TwinRL achieves near-**100%** success in both in-distribution and OOD regions, with **30%+** faster convergence than prior real-world RL methods using only **20 min** of on-robot interaction
  - Contributed to **Video2Act**, a dual-system VLA using a video diffusion model to guide a fast DiT action expert; participated in real-world experiments across 6 tasks, surpassing prior SOTA VLAs by **21.7%** in success rate

PROJECTS

- **DexTeleGym: Immersive VR Bimanual Dexterous Teleoperation in Isaac Gym [🐙]** *Dec. 2024 - Jan. 2025*
  Tools: Python, Isaac Gym, Meta Quest 3, WebRTC, Pinocchio, dex-retargeting
  - Developed an immersive VR teleoperation pipeline in NVIDIA Isaac Gym based on OpenTeleVision, enabling real-time control of dual Franka Panda arms with Inspire dexterous hands using Meta Quest 3
  - Implemented low-latency WebRTC streaming from simulation to the headset, with multi-modal visual feedback
  - Integrated real-time arm IK via Pink/Pinocchio and human-to-robot hand retargeting via dex-retargeting
  - Designed gesture-based controls for episode recording and display switching to support demonstration collection for robot learning; open-sourced with **60+** GitHub stars

HONORS AND AWARDS

- **National Scholarship (Top 0.2% nationwide)** *Dec. 2024*
- **Wuhan University Excellent Student Class A Scholarship (Top 5% university-wide)** *Oct. 2024*
- **Wuhan University Huang Zhangren Special Scholarship** *Oct. 2023*
- **First Prize, Provincial Level, Mathematics Competition of Chinese College Students** *Dec. 2023*
- **National Training Program of Innovation and Entrepreneurship for Undergraduates** *Mar. 2024*

SERVICES

- **Reviewer, IEEE International Conference on Robotics and Automation (ICRA)**
- **Reviewer, IEEE Robotics and Automation Letters (RA-L)**
- **Reviewer, IEEE International Conference on Multimedia and Expo (ICME)**